

SEMESTER END EXAMINATIONS - FEBRUARY 2024

Program	: B.E :- Information Science and Engineering	Semester	: V
Course Name	: Software Engineering	Max. Marks	: 100
Course Code	: IS51	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- With a neat diagram explain Activity model of the insulin pump. CO1 (06)
 - With a neat diagram explain the requirements engineering process. CO1 (06)
 - With a neat diagram explain waterfall model and identify its advantages and its limitations. CO1 (08)
- With a suitable use case explain and compare the Plan driven processes and agile processes with its advantages and limitations. CO1 (08)
 - With a neat diagram explain the different Phases in the Rational Unified Process. CO1 (06)
 - What is the need of extreme programming? Draw and describe the extreme programming cycle. CO1 (06)

UNIT - II

- Define and differentiate functional and nonfunctional requirements. CO2 (06)
 - What is the requirement specification? Explain ways of system requirements. CO2 (08)
 - What is requirement validation? List and explain any four different checks to be carried out during requirement validation Process? CO2 (06)
- Briefly discuss the important activities of requirements engineering process with a neat diagram. CO2 (06)
 - Explain requirement elicitation and analysis process. CO2 (06)
 - With the diagram , explain Ethnography and prototyping for requirements analysis. CO2 (08)

UNIT - III

- What is an architectural style? Illustrate with neat sketches, advantages and disadvantages of the repository and layered architectural design models. CO3 (10)
 - Draw three use case diagrams and three class diagrams for Library Management System. CO3 (10)
- Explain any five design principles used while designing class-based components. CO3 (10)
 - Discuss the software architecture of an ATM system. Illustrate the functions of the input, process and output components. CO3 (10)

UNIT- IV

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|----|----|--|-----|------|
| 7. | a) | Distinguish between verification and validation processes in software testing. | CO4 | (08) |
| | b) | What do you mean by unit testing in the context of object-oriented software development? Explain with an example. Identify the parts that would make an effective automated unit test. | CO4 | (06) |
| | c) | With the diagram, explain the general model of the reengineering process. | CO4 | (06) |
| 8. | a) | Illustrate any five differences between software test engineer and hardware test engineer. | CO4 | (10) |
| | b) | Describe various defects encountered at different phases of software life-cycle. | CO4 | (10) |

UNIT - V

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| 9. | a) | Draw and describe the project scheduling process. | CO5 | (10) |
| | b) | Give the need of measurement. Discuss what a good manager looks for measurement. | CO5 | (10) |
| 10. | a) | Explain why the process of project planning is iterative and why a plan must be continually reviewed during software project. | CO5 | (10) |
| | b) | Describe different process areas and goals in the CMMI. | CO5 | (10) |
